
MSML 605

Version Control

Introduction

- New changes keep coming up
- We need to keep track of the changes
- There is some mechanism to see what changed overtime and undo certain actions

System Administration

- New software is updated, may be server updates
 - If something breaks, without version control it would be a hassle.
 - With version control you could rollback to a stable version.
-

Keeping track

- Making copies of the work
 - Something that was removed only to be added later
 - Keeping historical copies is elementary version control (primitive).
 - Who did what and why is lost.
-

Comparing files

- Two copies of the same code from different times
 - Eyeball them?
 - *diff* command
 - *meld*, *kdiff3*, *vimdiff*
 - *patch*
-

Version Control

- Keeps track of all the versions
 - Helps retrieve past versions and who changed the files and when
 - Files are organized in repositories
 - A repository can have thousands of contributors.
-

Git



- Creator - Linus Torvalds
- Open source - originally created to manage development of Linux kernel
- Distributed architecture - copies on the local as well as the remote machines.
- Git can be used as a server or a client or both on a machine.

Git

- Clients can communicate with the server over http, ssh or git own protocol.
- Multi-platform support
- Public servers GitHub, gitlab, etc.
- Official Git website, <https://git-scm.com/> (scm - source control management)
- Other tools like Subversion (<https://subversion.apache.org/>), Mercurial (<https://www.mercurial-scm.org/>) etc.

Git - Installation

- *git* —*version*
to check if it is already installed
- *conda install git*
installation guide - (<https://git-scm.com/book/en/v2/Getting-Started-Installing-Git>)
download page - (<https://git-scm.com/downloads>)
- Windows comes with the environment *MinGW64* to operate like Linux
- There are some IDE's available

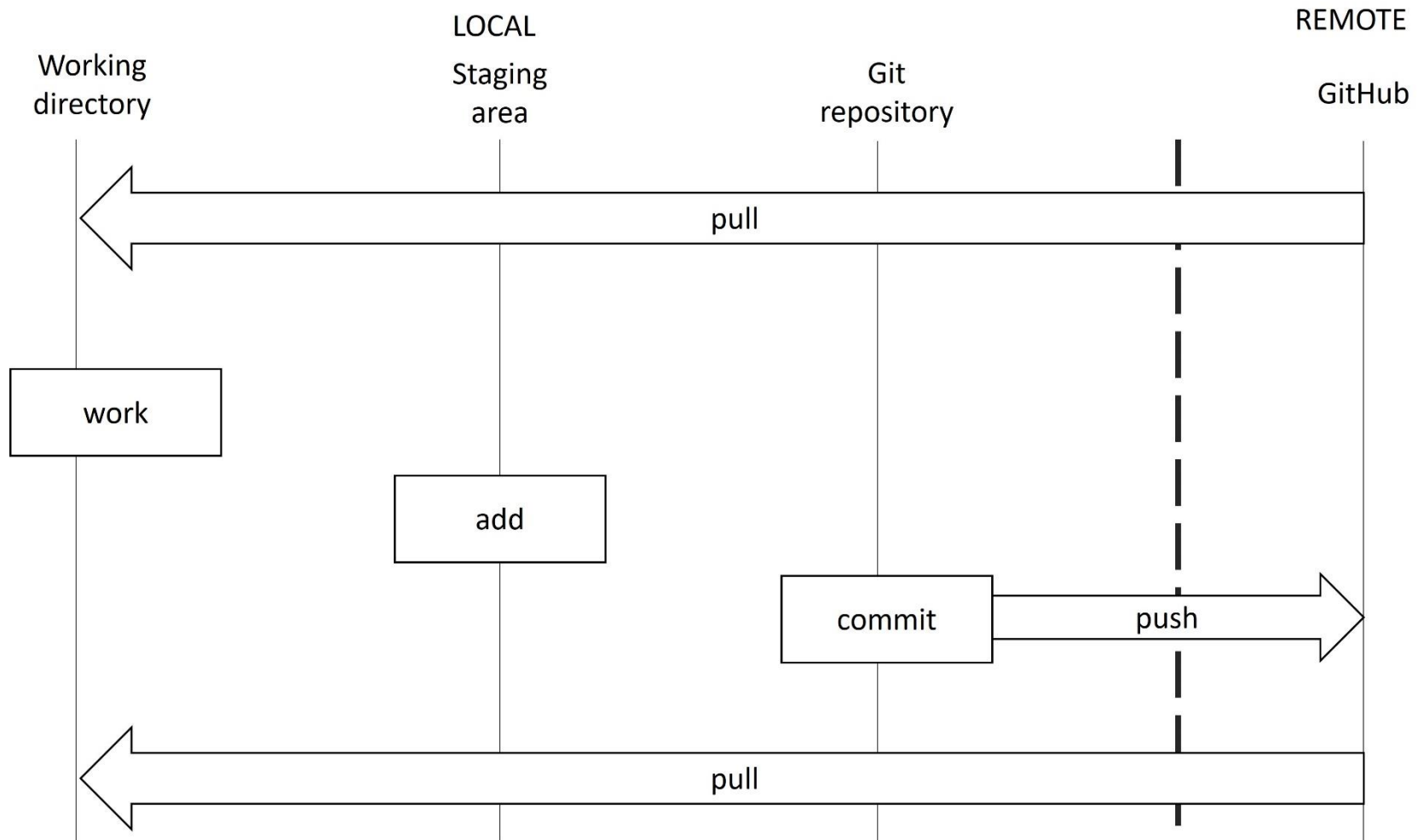
Git commands

- To keep track of who made the changes
git config —global user.email “email”
git config —global user.name “name”
- *User name and email* are global for all git repositories
- *Git init*
initializes a repository in the current directory

ls -la

ls -l

Git workflow



Git directory

- *git clone* creates a similar directory as *git init*
 - Area outside the Git directory is the working directory - current version of the project
 - Create a new file or copy an existing file.
 - Add it to the staging area.
git add <filename>
-

Staging area and commit

- Git directory, working tree and staging area
 - *git status*
 - *git commit*
 - File Stages: modified, staged, committed
-

Tracking

- Check status
 - Modify a file that is under version control
 - Check status again
 - Add the file to the staging area
 - Commit the changes
 - Check status
-

Git workflow

```
(base) Mohammads-MBP:repositories nayeem$ git config -l
http.sslverify=true
http.sslcapath=/anaconda3/ssl/cacert.pem
http.sslcainfo=/anaconda3/ssl/cacert.pem
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=[REDACTED]
user.email=[REDACTED]
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
```

- *git log*

No staging area

- Any changes to tracked files

git commit -a -m "message"

- HEAD alias represent the currently checked out snapshot of the repository
 - HEAD can be a commit in a different branch of a project
-

Git log

- *git log -p*

p for patches

- *Uses a paging tool*

- *git show*

- *git log —stat*

Git working area

- `git diff`
difference before commit
 - `git add -p`
 - `git diff —staged`
-

Removing and renaming files

- `Git rm <filename>`
 - `Git commit -m "message"`
 - `Git mv <old file name> <new file name>`
 - `Git commit -m "message"`
-

Ignore some files and remove changes

- Add them to .gitignore file
echo .abc >.gitignore

example: <https://gist.github.com/octocat/9257657>

- *git checkout <filename>*
to undo changes made that you don't want to add to the staging area.
-

remove changes in staged files

- *git restore —staged <filename>*
to undo changes made and added to the staging area.
- *git restore <filename>*

Wrong commit message

- *Create two files*
- *Add one file*
- *Commit one file with the message that two files were added*
- *Then add second file*
- *git commit —amend*
to fix the message

Don't use for public repositories

Wrong commit - undo last commit

- *git revert HEAD*
- *git log -p -2*

Wrong commit - by commit id

- *git log -1*

```
(msmlDockerApp) bash-3.2$ git log -1  
commit 0107f7c9e12fc1c8670d490d408b77ac73b59fc3 (HEAD -> master)
```

- *git log*

- *git show*

```
git show 398f4175772bc2058a42db7500b355262d3b2009
```

- *git revert*

```
git revert 398f4175772bc2058a42db7500b355262d3b2009
```

Branches

- *git branch*
command to list, create , delete and manipulate branches
- *git branch*
- *git branch new-development*
- *git checkout new-development*
- *git checkout -b new-branch*

Branches

- *git checkout -b new-branch*
- *Create a new file (list_numbers.py) , add and commit it*
- *git log -1*

```
commit 0527648028ff19a907adc9187ba8e031f65fe653 (HEAD -> new-branch)
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date:   Tue Apr 28 16:55:32 2020 -0400

    Added list_number to the new-branch
```

Branches

- *git status*

```
On branch new-branch
nothing to commit, working tree clean
```

- *git checkout master*

```
Switched to branch 'master'
```

- *git log -2*

- *ls -l*

```
-rwxr-xr-x  1 nayeem  staff   134 Apr 28 16:23 code.py
-rw-r--r--  1 nayeem  staff     0 Apr 16 17:49 file1.py
-rw-r--r--  1 nayeem  staff     0 Apr 16 17:49 file2.py
```

Branches

- *git branch*

```
(base) $ git branch
* master
  new-branch
  new-development
```

- *git branch -d new-development*

```
Deleted branch new-development (was daa034c).
```

- *git branch -d new-branch*

```
(base) $ git branch -d new-branch
error: The branch 'new-branch' is not fully merged.
If you are sure you want to delete it, run 'git branch -D new-branch'.
```

Merging

- *git branch*

```
* master  
new-branch
```

- *git merge new-branch*

```
Updating daa034c..0527648  
Fast-forward  
 list_number.py | 3 +++  
1 file changed, 3 insertions(+)  
create mode 100644 list_number.py
```

- *git merge*
uses fast forward and three-way merge

Merge Conflicts

- *git branch*

```
* master  
new-branch
```

- *Modify a file and commit to the master branch*

```
git commit -a -m "print line added"
```

- *Checkout a branch*

```
git checkout new-branch
```

Merge Conflicts

- *Modify the same file in the new branch and commit to it*

```
git commit -a -m "Added print \"Done\" to list_number"
```

- *Now Checkout the master branch*

```
git commit -a -m "Added print \"Done\" to list_number"
```

- *Now merge the branch*

```
git merge new-branch
```

```
Auto-merging list_number.py
CONFLICT (content): Merge conflict in list_number.py
Automatic merge failed; fix conflicts and then commit the result.
```

Merge Conflicts

- *Check the status. - git status*

```
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   list_number.py

no changes added to commit (use "git add" and/or "git commit -a")
```

- *See the file for the conflicts*

```
#!/anaconda3/bin/python
for i in range(10):
    print(i)

<<<<<<< HEAD
print(i)
=====
print("Done")
>>>>>>> new-branch
```

Merge Conflicts

- *Keep both changes and add the file*

```
git add list_number.py
```

- *Check the status*

```
git status
On branch master
All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

Changes to be committed:
  modified:   list_number.py
```

- *git commit*

Merge Conflicts

- *git commit*

```
Merge branch 'new-branch'

# Conflicts:
#   list_number.py
#
# It looks like you may be committing a merge.
# If this is not correct, please remove the file
#   .git/MERGE_HEAD
# and try again.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch master
# All conflicts fixed but you are still merging.
#
# Changes to be committed:
#   modified:   list_number.py
#
```

- *Add a comment like, Keeping changes from both branches*

Merge Conflicts

- *Check the commit history*

```
git log --graph --oneline
```

```
* d593708 (HEAD -> master) Merge branch 'new-branch'
| \
| * 5b822fd (new-branch) Added print Done to list_number
| * | a1262ab print line added
| /
* 0527648 Added list_number to the new-branch
* daa034c Revert "code.py renamed to py_code.py"
* 0107f7c Revert "Added nomsese" Roll back because of errors
* a02b528 Added nomsese
* 4d9883d Added two files file1 to verify rrsults file2 to display results
* 5b1e353 added .gitignore file
* 398f417 code.py renamed to py_code.py
* feldefdf removed test.py since it was empty and useless
* 2ede863 Added empty test.py file
* 8b4ce55 Added done at the end
* f9b2351 print pandas version
* df367a2 imported pandas library
* 74d16f6 first version of code.py file
```

- *git merge —abort*
to reset to the previous commit to throw away the changes.

Remote repository

- *Github*
- *Provides public and private repositories*
- *You will need an account. It is free.*

The screenshot shows the GitHub homepage. At the top is a dark navigation bar with the GitHub logo, a search bar, and links for Pull requests, Issues, Marketplace, and Explore. On the left, the 'Repositories' section lists several public repositories by user 'nayeemmmz'. The main content area features a large, light blue and green card titled 'Learn Git and GitHub without any code!' which provides a brief guide on how to get started. To the right of this card are two promotional banners: one for 'GitHub Satellite on May 6' and another for 'GitHub is now free for teams'.

Repositories New

Find a repository...

- [nayeemmmz/cmsc426Spring2...](#)
- [nayeemmmz/MSML605](#)
- [nayeemmmz/umd426.github.io](#)
- [nayeemmmz/umdcmssc426.git...](#)
- [nayeemmmz/cmsc320Summe...](#)
- [nayeemmmz/nayeemmmz](#)
- [ginawong/depth_estimation_...](#)

Learn Git and GitHub without any code!

Using the Hello World guide, you'll create a repository, start a branch, write comments, and open a pull request.

Join us for GitHub Satellite on May 6
GitHub's product and community event is completely virtual and free to join this year. Check out the schedule and add sessions to your calendar.

GitHub is now free for teams
GitHub Free gives teams private repositories with unlimited collaborators at no cost. GitHub Team is now reduced to \$4 per user/month.

Remote repository

■ *Create a repository - test*

Owner

 nayeemz ▾

 /

Repository name *

Great repository names are short and memorable. Need inspiration? How about **fuzzy-octo-engine**?

Description (optional)

☒  **Public**

Anyone can see this repository. You choose who can commit.

☐  **Private**

You choose who can see and commit to this repository.

Skip this step if you're importing an existing repository.

☐ **Initialize this repository with a README**
This will let you immediately clone the repository to your computer.

Add .gitignore: **None** ▾

 |

Add a license: **None** ▾



Create repository

Remote repository

■ *New repository*

 nayeemmmz / test Private

 Watch ▾

0

 Star

0

 Fork

0

<> Code

! Issues 0

 Pull requests 0

 Actions

 Projects 0

 Security 0

 Insights

 Settings

No description, website, or topics provided.

Edit

[Manage topics](#)

https://github.com/nayeemmmz/test.git

 1 commit

 1 branch

 0 packages

 0 releases

Branch: master ▾

New pull request

Create new file

Upload files

Find file

Clone or download ▾

 nayeemmmz Initial commit

Latest commit 2ecc9ac 16 seconds ago

 README.md

Initial commit

16 seconds ago

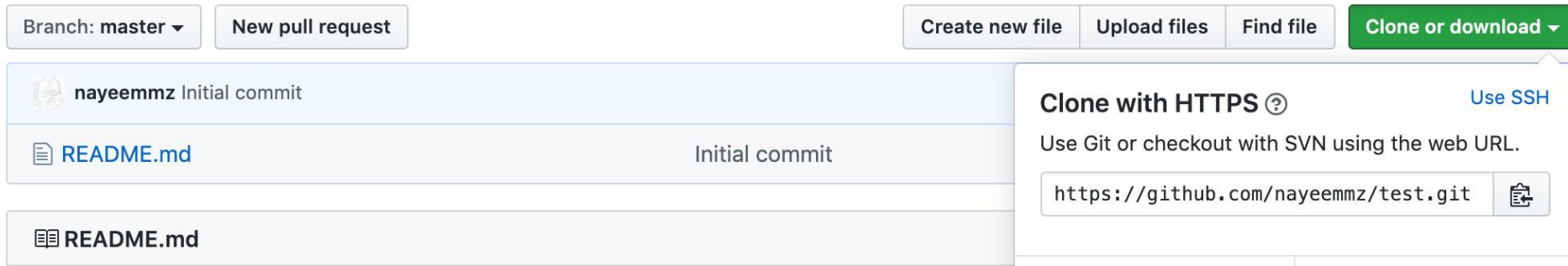
 README.md



test

Clone - Remote repository

■ *Clone a repository*



The screenshot shows a GitHub repository interface. At the top, there are buttons for 'Branch: master', 'New pull request', 'Create new file', 'Upload files', 'Find file', and a green 'Clone or download' button. Below these, the repository name 'nayeemmz' and 'Initial commit' are visible. A file named 'README.md' is listed. A modal dialog titled 'Clone with HTTPS' is open, showing the URL 'https://github.com/nayeemmz/test.git' and a 'Use SSH' link.

```
git clone https://github.com/nayeemmz/test.git
```

```
Cloning into 'test'...
Username for 'https://github.com':
Password for 'https://nayeemmz@github.com':
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), done.
```

■ *A directory is created locally*

Update local repository

- *Go to local test repository and update README file*
- *Commit the changes*

```
git commit -a -m "Added two lines to README.md file"
```

- *Push the changes to the remote repository*

```
git push
```

```
Enumerating objects: 5, done.  
Counting objects: 100% (5/5), done.  
Delta compression using up to 4 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0)  
To https://github.com/nayeemmz/test.git  
2ecc9ac..e2da63b master -> master
```

Check the remote repository

■ *remote repository*

 2 commits

 1 branch

 0 packages

 0 releases

Branch: master ▾

New pull request

Create new file

Upload files

Find file

Clone or download ▾

 nayeemmz Added two lines to README.md file Latest commit e2da63b 4 minutes ago

 [README.md](#) Added two lines to README.md file 4 minutes ago

 README.md 

 **test**

This repo is our first repository. We will populate it with fancy ML scripts.

Avoid Authentication

- *Use ssh or*
- *Credential helper -*

```
git config --global credential.helper cache
```

- *Pull the remote repository*

```
git pull
```

```
Username for 'https://github.com':
```

```
Password for 'https://nayeemmmz@github.com':
```

```
Already up to date.
```

- *Pull the remote repository again*

```
git pull
```

```
Already up to date.
```

Remote repository

- *Use ssh or https*
- *Check the remote url*

```
git remote -v
```

```
origin https://github.com/nayeemz/test.git (fetch)  
origin https://github.com/nayeemz/test.git (push)
```

- *Show origin on remote computer*

```
origin https://github.com/nayeemz/test.git (fetch)  
origin https://github.com/nayeemz/test.git (push)
```

```
* remote origin  
Fetch URL: https://github.com/nayeemz/test.git  
Push URL: https://github.com/nayeemz/test.git  
HEAD branch: master  
Remote branch:  
  master tracked  
Local branch configured for 'git pull':  
  master merges with remote master  
Local ref configured for 'git push':  
  master pushes to master (up to date)
```

Remote repository

- *Remote branches*

```
git branch -r
```

```
origin/HEAD -> origin/master  
origin/master
```

- *Check status*

```
git status
```

```
On branch master  
Your branch is up to date with 'origin/master'.  
  
nothing to commit, working tree clean
```

Remote repository

■ *Update an commit through Github*

 **nayeemz** Update README.md 53ec7d8 2 minutes ago

1 contributor

6 lines (4 sloc) | 106 Bytes

Raw Blame History

test

This repo is our first repository. We will populate it with fancy ML scripts.

Updating on Github

■ *Check remote repository locally*

```
git remote show origin
```

```
* remote origin
Fetch URL: https://github.com/nayeemz/test.git
Push URL: https://github.com/nayeemz/test.git
HEAD branch: master
Remote branch:
  master tracked
Local branch configured for 'git pull':
  master merges with remote master
Local ref configured for 'git push':
  master pushes to master (local out of date)
```

Syncing

- *Sync remote and local repositories*

```
git fetch
```

```
remote: Enumerating objects: 5, done.  
remote: Counting objects: 100% (5/5), done.  
remote: Compressing objects: 100% (2/2), done.  
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0  
Unpacking objects: 100% (3/3), done.  
From https://github.com/nayeemmh/test  
   e2da63b..53ec7d8  master    -> origin/master
```

Remote log

- *Check the log of the remote repository*

```
git log origin/master
commit 53ec7d8746f15ac51c9388e9ae40d9477bfbf35f (origin/master, origin/HEAD)
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date: Thu Apr 30 15:52:15 2020 -0400

    Update README.md

commit e2da63b4f06afd7a40747b704a4088b5dc93bf62 (HEAD -> master)
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date: Wed Apr 29 16:07:06 2020 -0400

    Added two lines to README.md file

commit 2ecc9ac4606ace3e394bdd4ce0c8e0bf48f8e556
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date: Wed Apr 29 15:37:15 2020 -0400

    Initial commit
```

Missing commit

- *Check status*

```
git status
```

```
On branch master
Your branch is behind 'origin/master' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
```

- *Merge the changes*

```
git merge origin/master
```

```
Updating e2da63b..53ec7d8
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
```

Check the log again

■ *Check status*

```
git log
```

```
commit 53ec7d8746f15ac51c9388e9ae40d9477bfbf35f (HEAD -> master, origin/master, origin/HEAD)  
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>  
Date: Thu Apr 30 15:52:15 2020 -0400
```

Update README.md

```
commit e2da63b4f06afd7a40747b704a4088b5dc93bf62  
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>  
Date: Wed Apr 29 16:07:06 2020 -0400
```

Added two lines to README.md file

```
commit 2ecc9ac4606ace3e394bdd4ce0c8e0bf48f8e556  
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>  
Date: Wed Apr 29 15:37:15 2020 -0400
```

Initial commit

Conflicts

- *Update a file locally*
- *Check the changes to accept*

```
git add -p
```

```
diff --git a/README.md b/README.md
index 6f48ba6..59d7539 100644
--- a/README.md
+++ b/README.md
@@ -4,3 +4,5 @@ This repo is our first repository.
    We will populate it with fancy ML scripts.

    Updating on Github
+
+New changes for conflicts
Stage this hunk [y,n,q,a,d,e,?]? y
```

- *Commit*

```
git commit -m "Added a line to test conflicts"
```

Conflicts

- *Update the file in the remote repository on Github*

8 lines (5 sloc) | 147 Bytes

Raw

Blame

History



test

This repo is our first repository. We will populate it with fancy ML scripts.

Updating on Github

Added text to test conflicts on Github

- *Push the change you made locally*

`git push`

```
Username for 'https://github.com': nayeemmmz
Password for 'https://nayeemmmz@github.com':
To https://github.com/nayeemmmz/test.git
 ! [rejected]          master -> master (fetch first)
error: failed to push some refs to 'https://github.com/nayeemmmz/test.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

Conflicts

■ Merge first

```
git pull
```

```
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), done.
From https://github.com/nayeemz/test
   53ec7d8..e81cc08  master    -> origin/master
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
```

■ Check log

```
git log --graph --oneline --all
```

```
* e81cc08 (origin/master, origin/HEAD) Update README.md
| * bd79b98 (HEAD -> master) Added a line to test conflicts
|/
* 53ec7d8 Update README.md
* e2da63b Added two lines to README.md file
* 2ecc9ac Initial commit
```

Conflicts

■ Check the commits on origin/master

```
git log -p origin/master
```

```
commit e81cc08dcee9bb6df20086acfc9f4b327d2ffd07 (origin/master, origin/HEAD)
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date: Thu Apr 30 16:40:31 2020 -0400

    Update README.md

diff --git a/README.md b/README.md
index 6f48ba6..d163ffc 100644
--- a/README.md
+++ b/README.md
@@ -4,3 +4,5 @@ This repo is our first repository.
    We will populate it with fancy ML scripts.

    Updating on Github
+
+Added text to test conflicts on Github

commit 53ec7d8746f15ac51c9388e9ae40d9477bfbf35f
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date: Thu Apr 30 15:52:15 2020 -0400

    Update README.md

diff --git a/README.md b/README.md
index 472187a..6f48ba6 100644
--- a/README.md
+++ b/README.md
@@ -2,3 +2,5 @@
    This repo is our first repository.
    We will populate it with fancy ML scripts.
+
+Updating on Github

commit e2da63b4f06afd7a40747b704a4088b5dc93bf62
Author: Mohammad Nayeem Teli <mohammad.nayeem@gmail.com>
Date: Wed Apr 29 16:07:06 2020 -0400

    Added two lines to README.md file

diff --git a/README.md b/README.md
index 00bcb6e..472187a 100644
--- a/README.md
+++ b/README.md
@@ -1,4 @@
```

Conflicts

- *Correct or merge the multiple changes in the file*

- *Add the file*

```
vi README.md  
git add README.md
```

- *Commit the changes*

```
git commit -m "Merged the changes from local and remote"
```

- *Push the changes*

```
(base) Mohammads-MBP:test nayeem$ git push  
Username for 'https://github.com': nayeemnz  
Password for 'https://nayeemnz@github.com':  
Enumerating objects: 10, done.  
Counting objects: 100% (10/10), done.  
Delta compression using up to 4 threads  
Compressing objects: 100% (4/4), done.  
Writing objects: 100% (6/6), 594 bytes | 297.00 KiB/s, done.  
Total 6 (delta 2), reused 0 (delta 0)  
remote: Resolving deltas: 100% (2/2), completed with 1 local object.  
To https://github.com/nayeemnz/test.git  
e81cc08..71020b5 master -> master
```

Conflicts

- *Check the log to verify*

```
(base) Mohammads-MBP:test nayeem$ git log --onegraph --oneline
fatal: unrecognized argument: --onegraph
(base) Mohammads-MBP:test nayeem$ git log --graph --oneline
*    71020b5 (HEAD -> master, origin/master, origin/HEAD) Merged the changes from local and remote
| \
| * e81cc08 Update README.md
* | bd79b98 Added a line to test conflicts
| /
* 53ec7d8 Update README.md
* e2da63b Added two lines to README.md file
* 2ecc9ac Initial commit
```